AI-ASSISTED CODING
Assignment - 21.1
2303A510B4
Batch - 14

Lab 21 – Collaborative AI Coding: Pair Programming and Version Control Integration

TASK 1 – AI-Assisted Pair Programming

Prompt:

Generate a simple To-Do List Manager in Python with basic features:

- Add task
- View task
- Delete task

Initial Code:

```
tasks = []
while True:
    print("\n1. Add Task\n2. View Tasks\n3. Delete Task\n4. Exit")
    choice = int(input("Enter choice: "))
    if choice == 1:
        task = input("Enter task: ")
        tasks.append(task)
    elif choice == 2:
        for i in range(len(tasks)):
            print(i+1, tasks[i])
    elif choice == 3:
        num = int(input("Enter task number to delete: "))
        tasks.pop(num-1)
    elif choice == 4:
        break
```

Enhancement Prompt:

Improve the To-Do List program by adding:

- Error handling
- Input validation
- Mark task as completed

Improved Code

```
tasks = []
while True:
    print("\n1. Add Task\n2. View Tasks\n3. Delete Task\n4. Mark
Complete\n5. Exit")
    try:
        choice = int(input("Enter choice: "))
    except:
        print("Invalid input!")
        continue
    if choice == 1:
        task = input("Enter task: ")
        tasks.append({"task": task, "done": False})
    elif choice == 2:
        if len(tasks) == 0:
            print("No tasks available")
        else:
            for i in range(len(tasks)):
                status = "✔" if tasks[i]["done"] else "✗"
                print(i+1, tasks[i]["task"], "-", status)
    elif choice == 3:
        try:
            num = int(input("Enter task number: "))
            tasks.pop(num-1)
        except:
            print("Invalid index")
    elif choice == 4:
        try:
            num = int(input("Enter task number: "))
            tasks[num-1]["done"] = True
        except:
            print("Invalid index")
```

```python
    elif choice == 5:
        break
```

Output:

```
1. Add Task
2. View Tasks
3. Delete Task
4. Mark Complete
5. Exit
Enter choice: 1
Enter task: Study AI

1. Add Task
2. View Tasks
3. Delete Task
4. Mark Complete
5. Exit
Enter choice: 2
1 Study AI - ✗

1. Add Task
2. View Tasks
3. Delete Task
4. Mark Complete
5. Exit
Enter choice: 4
Enter task number: 1

1. Add Task
2. View Tasks
3. Delete Task
4. Mark Complete
5. Exit
Enter choice: 2
1 Study AI - ✓

1. Add Task
2. View Tasks
3. Delete Task
4. Mark Complete
5. Exit
Enter choice: 5
```

TASK 2 – Version Control Integration

Prompt:

Give Git commands to:

- Initialize repository
- Create branch
- Commit changes
- Merge branch

Code:

```
#Give Git commands to:
#- Initialize repository
#- Create branch
#- Commit changes
#- Merge branch
from heapq import merge
from mimetypes import init

git init
# Create branch
git checkout -b new-feature
# Commit changes
git add .
git commit -m "Add new feature"
# Merge branch
git checkout main
git merge new-feature
```

Output:

```
Initialized empty Git repository
Switched to branch 'feature-update'
[feature-update] commit successful
Merged successfully into main branch
```

TASK 3 – AI for Commit Message Generation

Prompt:

Generate a concise Git commit message for:
Added error handling and task completion feature

AI-Generated Message:

```
#Generate a concise Git commit message for:
#"Added error handling and task completion feature"

"Add error handling and task completion support"
```

Human Message:

Added new features and fixed errors

Comparison:

- AI message → Clear, structured, professional
- Human message → Vague

TASK 4 – Pair Programming Simulation

Student A Code:

```
num1 = int(input("Enter number 1: "))
num2 = int(input("Enter number 2: "))
print("Sum =", num1 + num2)
```

Prompt (Student B):

Improve calculator program with:

- Multiple operations
- Menu-driven system

Code:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
while True:
    print("1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit")
    ch = int(input("Enter choice: "))
    if ch == 1:
        print("Result:", a + b)
    elif ch == 2:
        print("Result:", a - b)
    elif ch == 3:
        print("Result:", a * b)
    elif ch == 4:
        if b != 0:
            print("Result:", a / b)
        else:
            print("Cannot divide by zero")
    elif ch == 5:
        break
```

Output:

```
Enter first number: 4
Enter second number: 2
1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit
Enter choice: 1
Result: 6
1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit
Enter choice: 2
Result: 2
1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit
Enter choice: 3
Result: 8
1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit
Enter choice: 4
Result: 2.0
1.Add 2.Subtract 3.Multiply 4.Divide 5.Exit
Enter choice: 5
```

TASK 5 – AI-Assisted Debugging

Buggy Code:

```
a = int(input("Enter number: "))
b = int(input("Enter number: "))
print("Division =", a / b)
```

Prompt:

Fix division by zero error in the code

Code:

```
a = int(input("Enter number: "))
b = int(input("Enter number: "))
if b == 0:
    print("Error: Division by zero not allowed")
else:
    print("Division =", a / b)
```

Output:

```
Enter number: 4
Enter number: 0
Error: Division by zero not allowed
```

```
Enter number: 10
Enter number: 5
Division = 2.0
```

TASK 6 – Feature Expansion using AI

Prompt:

Create a calculator with:

- History tracking
- Input validation

Code:

```
history = []
while True:
    print("\n1.Add 2.Subtract 3.Exit")
    try:
        ch = int(input("Enter choice: "))
    except:
        print("Invalid input")
        continue
    if ch == 3:
        break
    try:
        a = int(input("Enter first number: "))
        b = int(input("Enter second number: "))
    except:
        print("Invalid numbers")
        continue
    if ch == 1:
        result = a + b
    elif ch == 2:
        result = a - b
    else:
        print("Invalid choice")
        continue
    history.append(str(a) + " op " + str(b) + " = " + str(result))
    print("Result:", result)
print("\nHistory:")
for i in range(len(history)):
    print(history[i])
```

Output:

```
1.Add 2.Subtract 3.Exit
Enter choice: 1
Enter first number: 4
Enter second number: 2
Result: 6

1.Add 2.Subtract 3.Exit
Enter choice: 2
Enter first number: 5
Enter second number: 3
Result: 2

1.Add 2.Subtract 3.Exit
Enter choice: 3

History:
4 op 2 = 6
5 op 3 = 2
```